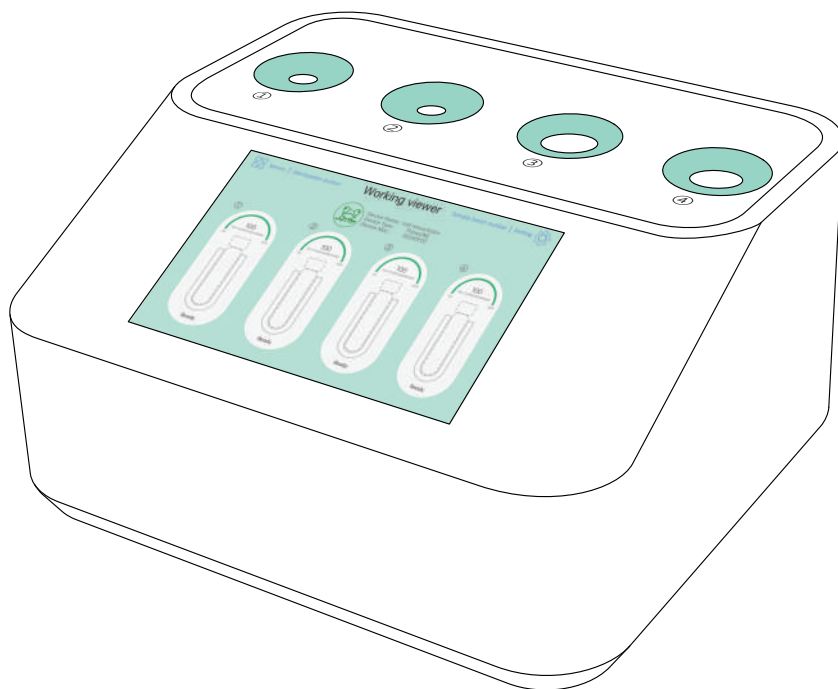


ThawLINE



User Manual

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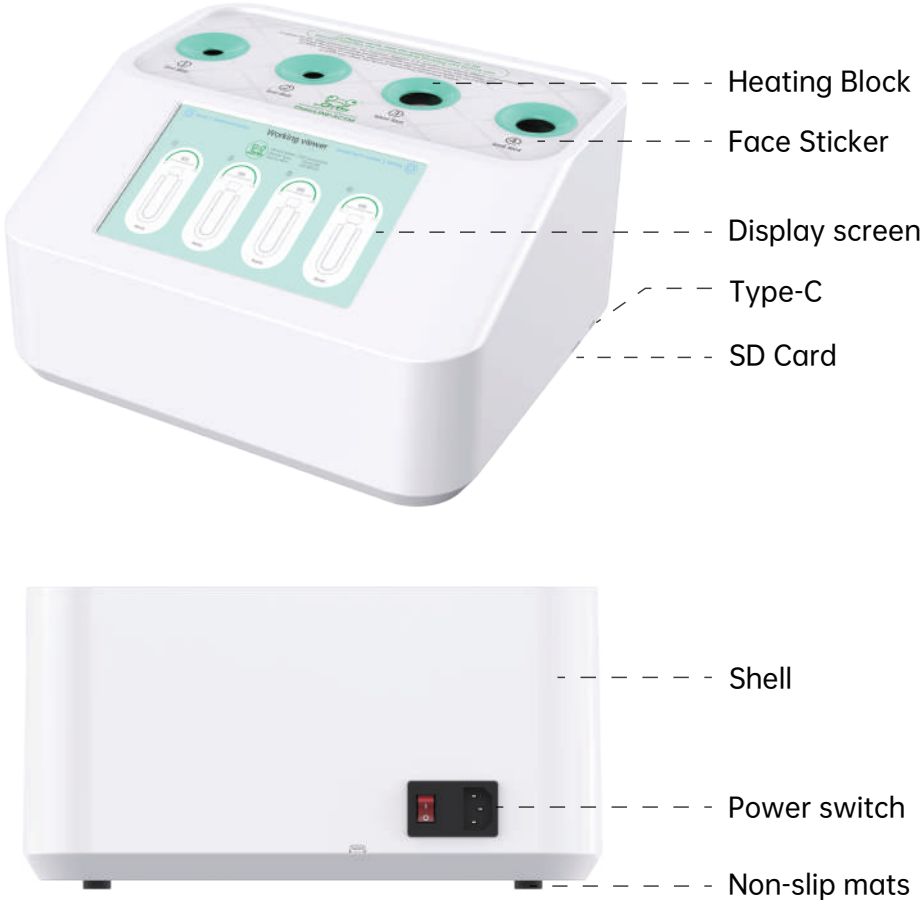
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Function description and appearance

The frozen storage, recovery and freezing operation of samples have been paid more and more attention by researchers. As an indispensable step in biological experiments, their stability and contamination rate will directly affect the results of experiments. Therefore, a good set of products can better help researchers do experiments, and can get faster and better experimental results. Similarly, industrial users are also more in need of similar products to stabilize production and improve efficiency, Kemesser has focused on such research and development for ten years, launched a series of products can be a good solution to biological samples frozen storage, recovery, freezing operations and other processes, to provide customers with a better and more stable operating environment, cell recovery instrument for efficient thawing of a variety of specifications of containers of living cells. After the biological test tube is directly removed from liquid nitrogen or dry ice, it is inserted into the slot of the cell resuscitate instrument. The thawing system can quickly restore the biological vitality of the cells through sensor technology and temperature control technology to achieve the effect of recovery.

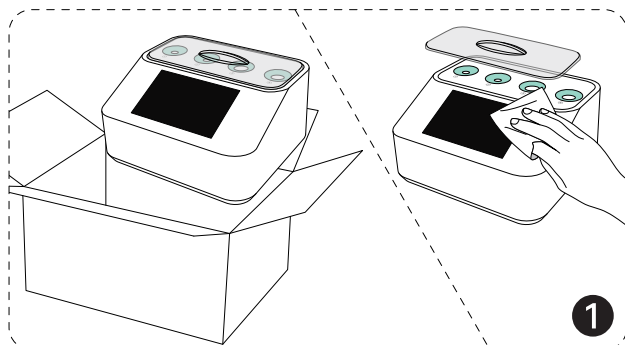


Suggested Settings

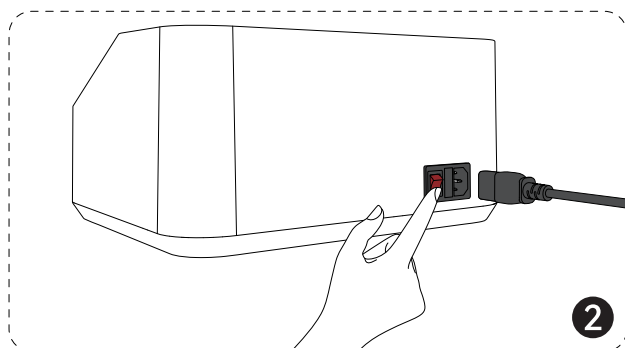
Note: Please set the parameters according to the volume of reagent in the container. The table below is for reference only.

Container / Cryopreservation Environment	Ultra-low Temperature Freezer	Liquid Nitrogen
T2 (2ml Cryogenic Tube)	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 60℃
T5 (5ml Cryogenic Tube)	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 60℃
A1 (1ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 60℃
A2 (2ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 60℃
A6 (6ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 60℃
A10 (10ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings: 120~140 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 120~140 Preheating Temperature Settings: 60℃
C2 (2ml Glass Vial)	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 40~70 Preheating Temperature Settings: 60℃
C6 (6ml Glass Vial)	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 70~80 Preheating Temperature Settings: 60℃
C10 (10ml Glass Vial)	Ice Crystal Parameter Settings: 140~160 Preheating Temperature Settings: 58℃	Ice Crystal Parameter Settings: 140~160 Preheating Temperature Settings: 60℃

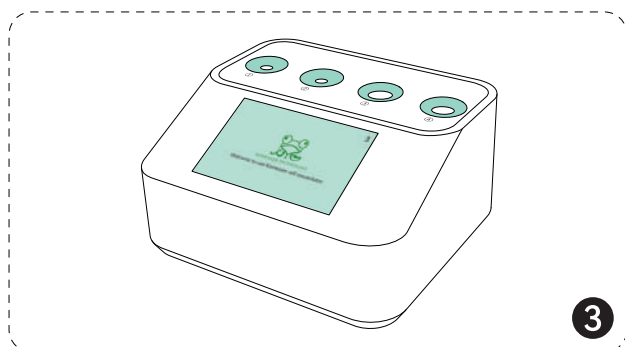
Quick Start Guide



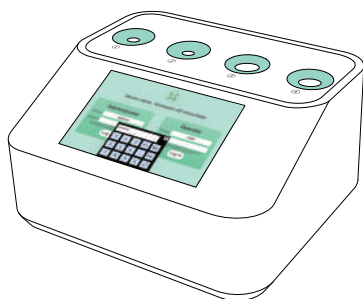
Remove the instrument from the carton, take off the dust cover, and wipe the surface of the device;



Carefully connect the power cord from the accessories to the power socket on the back of the device, then turn on the power switch;

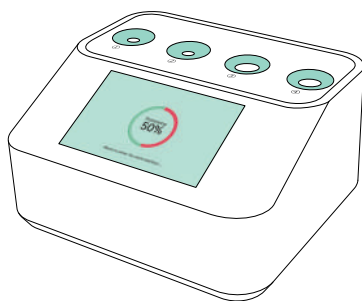


Enter the welcome screen;



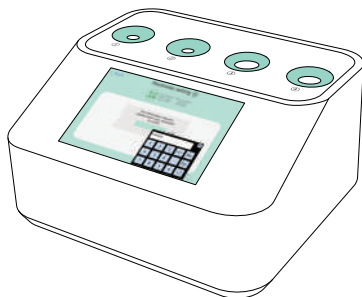
Enter the password as
an administrator;

4



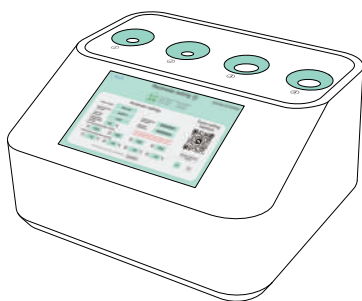
Wait for preheating;

5



After preheating is complete
and the main interface
appears, click **【Settings】**
in the top right corner and
enter the password;

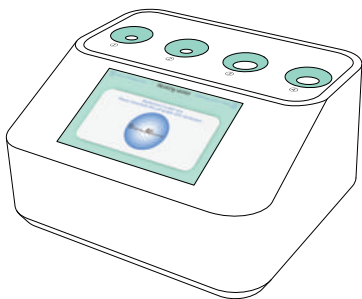
6



Enter Settings, adjust the ice crystal parameters and other settings, then click Confirm to save the data;

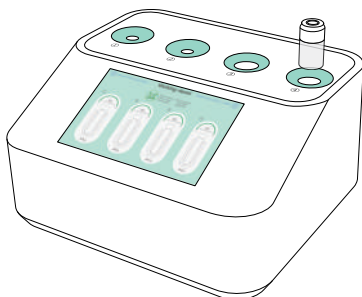
7

Note: Ice crystal parameters are a critical factor in the thawing process. Since each user's sample volume and cryopreservation solution may vary, we strongly recommend performing trial thawing with the same type and volume of cryopreservation solution before actual use to determine the optimal ice crystal settings. The ideal thawing state is achieved when 5%-15% of ice crystals remain at the end of the process.



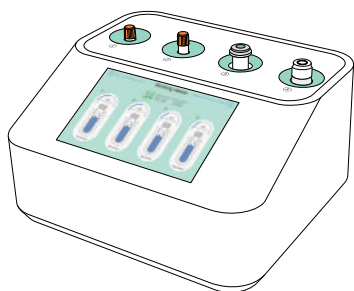
Return to the main interface, click **【Sterilization button】** in the top left corner to enter the sterilization screen, and wait for the process to complete;

8



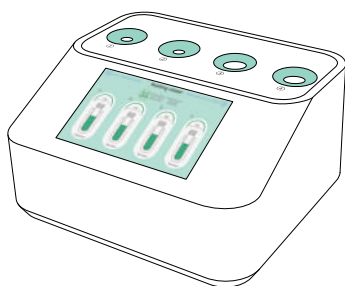
Return to the main interface, insert the sample tube, and begin cell thawing;

9



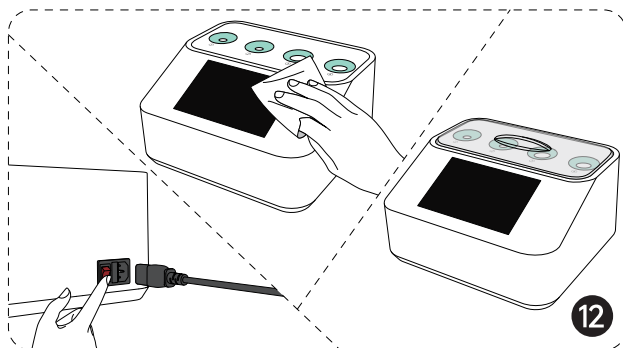
Enter the thawing screen,
cell thawing is in progress;

10



Remove the sample tube
promptly after cell thawing
is complete;

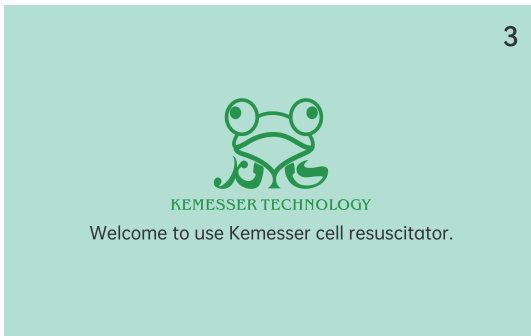
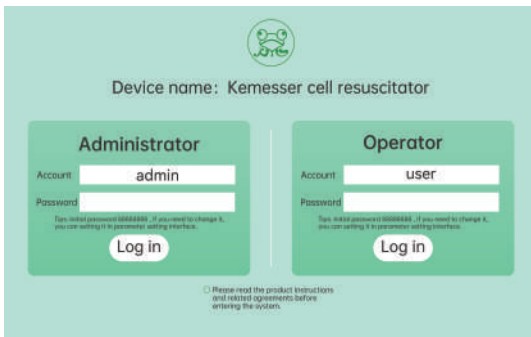
11

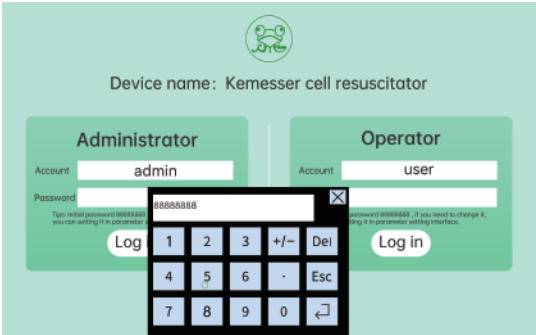


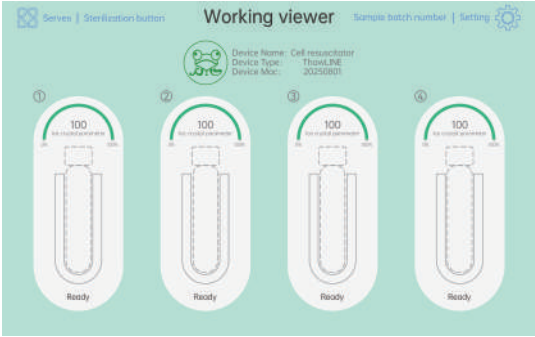
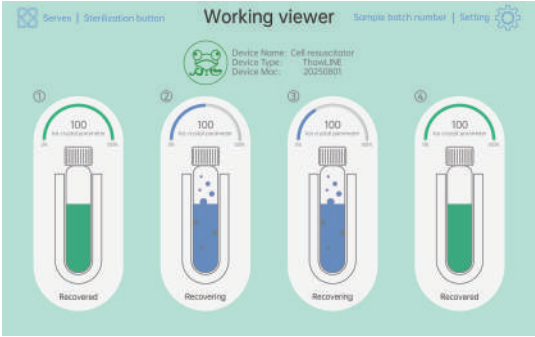
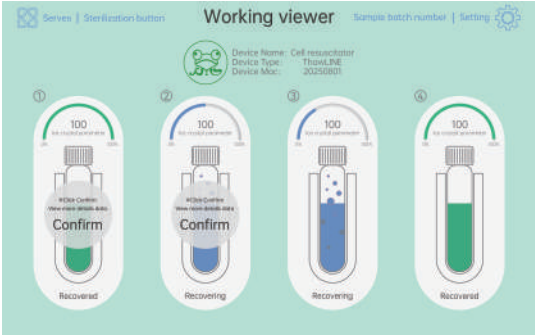
After placing the sample
tube, switch off the power
switch on the back of the
instrument and carefully
unplug the power cord.
Wipe the surface of the
instrument and then
replace the dust cover.

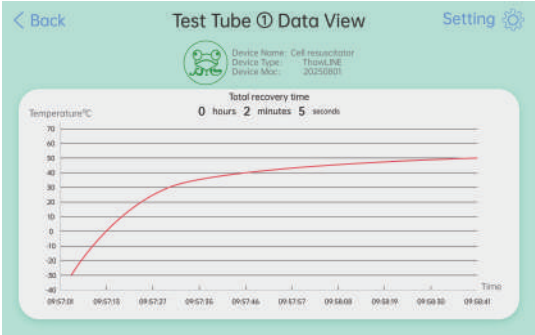
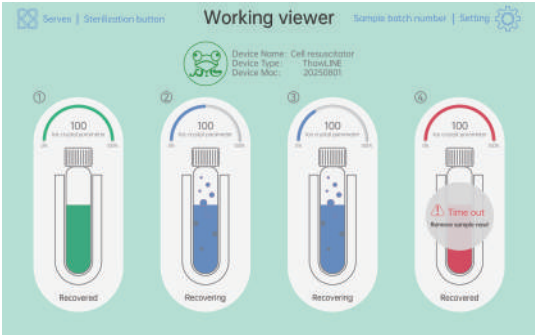
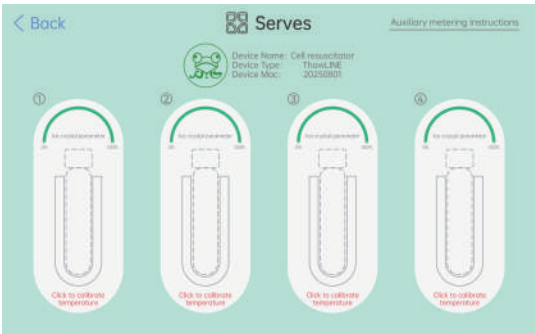
12

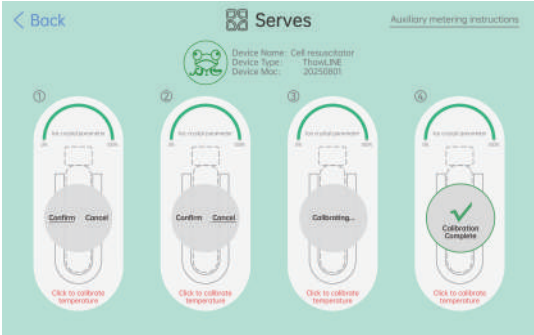
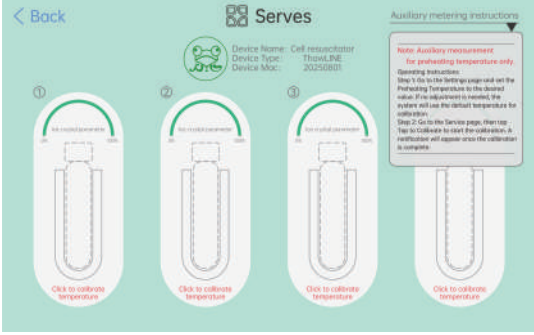
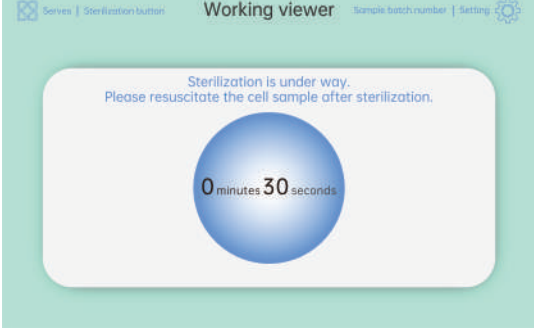
Software Interface

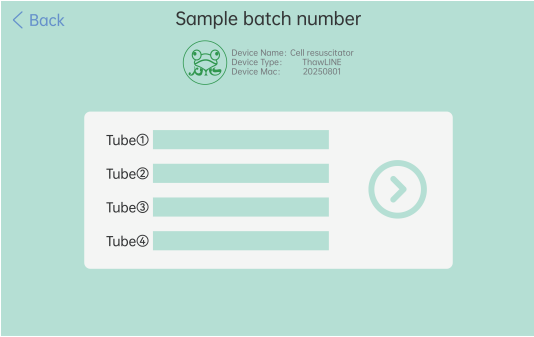
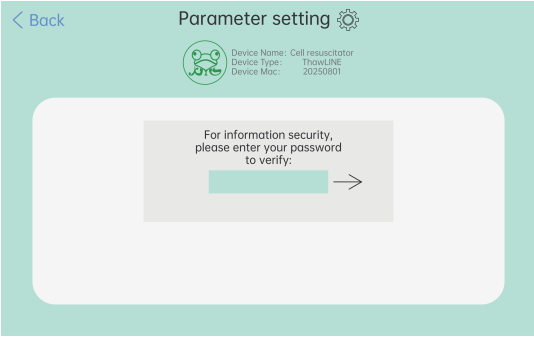
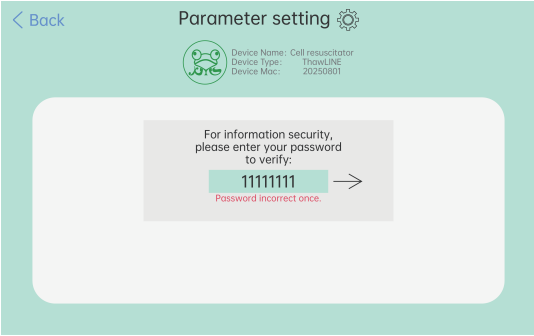
No.	Steps	Display Screen Indication	Sound
1	Check that the instrument is in good condition and connect the power supply;	N/A	N/A
2	Welcome Screen With normal power supply, toggle the switch on the back of the instrument. The display screen will light up automatically, as shown on the right;		N/A
3	Login Screen Enter the device password input screen. The default administrator account is: admin, the default operator account is: user, the default password for both accounts is: 88888888;		N/A

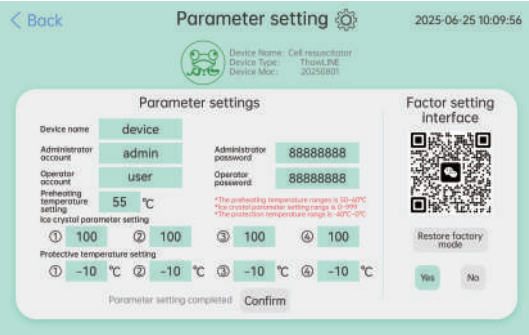
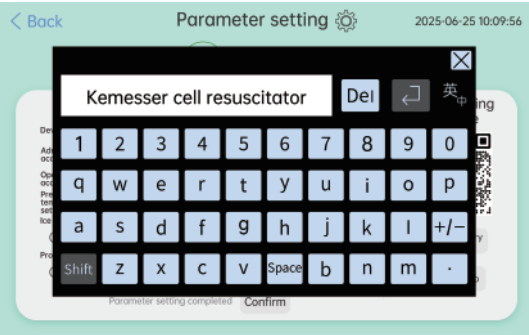
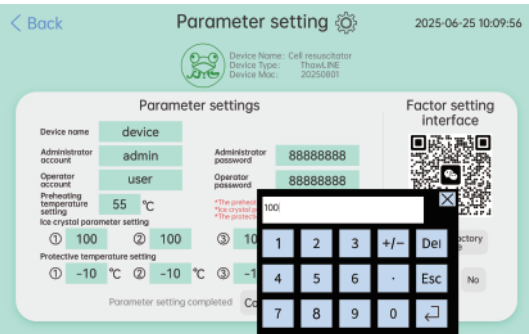
No.	Steps	Display Screen Indication	Sound
4	Incorrect Password If the device password is entered incorrectly 3 times, the screen on the right will appear. To retain experimental data and parameters, please contact Kemesser technical support. If you do not wish to save the data, you may choose the One-Key Factory Reset option. After the factory reset, all accounts and passwords will return to their default settings.		N/A
5	Login Screen Click the password input box, enter the password, press the Enter key on the keyboard, then click the Login button to proceed.		N/A
6	Preheating Screen Enter the preheating process: the red bar indicates the preheating progress, with 50% showing halfway completion. When it reaches 100%, the system will enter the main thawing interface.		N/A

No.	Steps	Display Screen Indication	Sound
7	Main Interface The main interface includes four sample tubes labeled 1/2/3/4. The ice crystal parameter represents the size of the ice crystals upon thawing completion (the larger the parameter value, the smaller the ice crystals).	 The screenshot shows the 'Working viewer' interface with a top bar containing 'Services', 'Sterilization button', 'Sample batch number', and 'Setting'. Below the bar, there's a status icon and text: 'Device Name: Cell resuscitator', 'Device Type: ThawLINE', and 'Device Mod.: 20250901'. The main area displays four sample tubes labeled 1, 2, 3, and 4. Each tube has a green progress bar at 100% and is labeled 'Ready' at the bottom. The tubes contain clear liquid.	N/A
8	Operation Interface Insert the sample tube into the instrument. Tubes labeled "Thawing" will appear in blue, and those labeled "Thawing Complete" will appear in green. The interface is shown in the image on the right.	 The screenshot shows the 'Working viewer' interface with the same top bar. The status text remains the same. The four sample tubes are shown with varying levels of liquid. Tubes 1 and 4 are labeled 'Recovered' and contain green liquid. Tubes 2 and 3 are labeled 'Recovering' and contain blue liquid. Each tube has a progress bar at 100%.	N/A
9	View Tube Data Click on a tube to display the "View Tube Details" button. Click Confirm to enter the tube data dashboard.	 The screenshot shows the 'Working viewer' interface with the same top bar. The status text remains the same. The four sample tubes are shown with varying levels of liquid. Tubes 1 and 4 are labeled 'Recovered' and contain green liquid. Tubes 2 and 3 are labeled 'Recovering' and contain blue liquid. Each tube has a progress bar at 100%. A 'Confirm' button is overlaid on each tube, with the text 'Click Confirm View tube details data' above it.	N/A

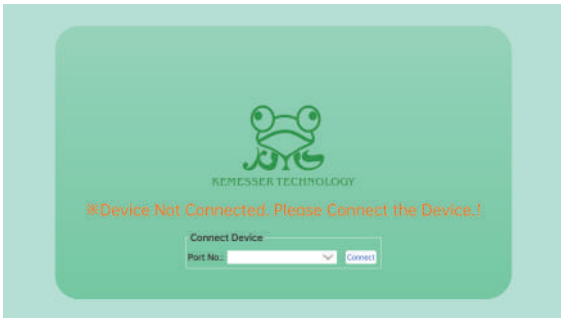
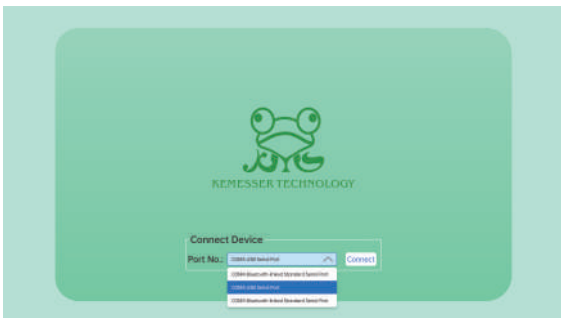
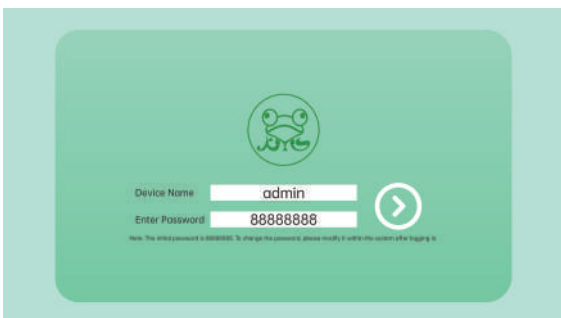
No.	Steps	Display Screen Indication	Sound
10	Tube Data Dashboard The screen, as shown on the right, displays the detailed data curves of the tube.		N/A
11	Main Interface If the tube is not removed in time, it will appear in red. The interface is shown on the right.		N/A
12	Service Interface Main Functions of the Service Interface: • Calibrate the parameters of the heating elements. • Measurement: After calibration, a standard instrument can be used to verify the accuracy of the device (e.g., if the heating element is set to 55°C, the standard instrument can measure whether the actual temperature matches).		N/A

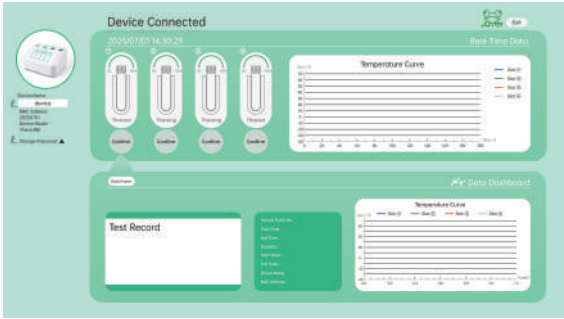
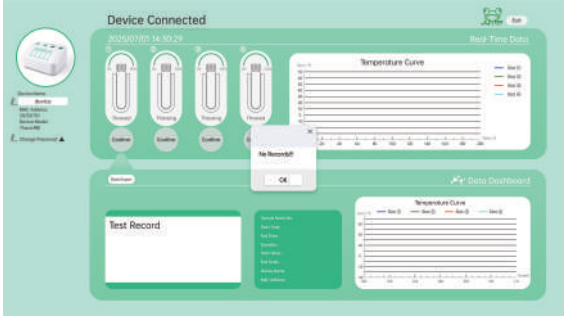
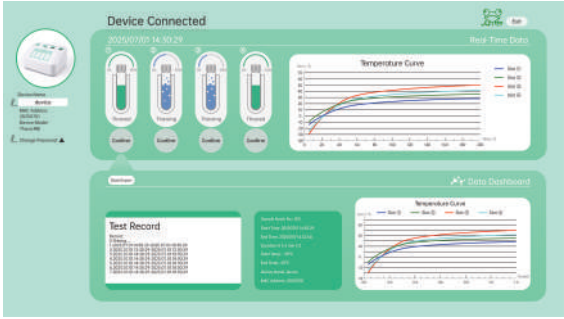
No.	Steps	Display Screen Indication	Sound
13	Temperature Calibration Click Calibration to enter the confirmation interface. After confirmation, the instrument will enter the calibration process until the temperature calibration of all four heating elements is completed. (For example: when the set temperature is changed from 45 °C to 55 °C, enter the Service menu and click Calibration. Calibration is considered complete only when the temperatures of all four heating elements stabilize at approximately 55 °C.)		N/A
14	Auxiliary metering instructions Click "Auxiliary Measurement" to display the auxiliary measurement instructions.		N/A
15	Sterilization Interface Click "Sterilization button" to start the ozone function with a 30-second countdown. After sterilization is complete, the system will return to the main working interface. During sterilization, the interface is locked and cannot be operated unless the device is powered off and restarted.		N/A

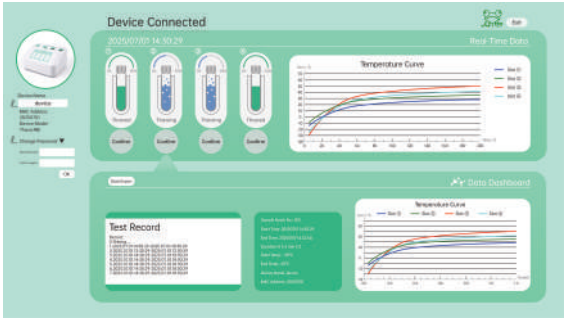
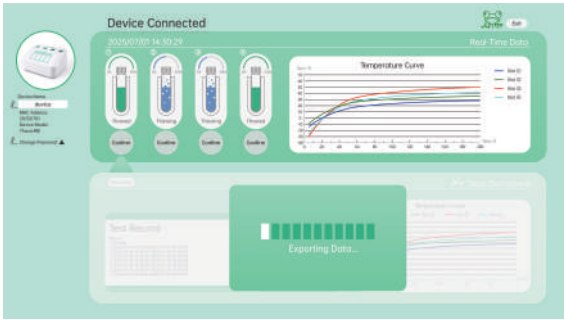
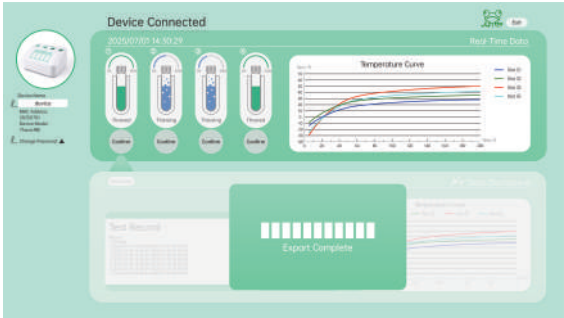
No.	Steps	Display Screen Indication	Sound
16	Sample Batch Number Interface Each test tube's sample batch number can be set individually. After entering the batch number, click the Save button to store the sample batch number.		N/A
17	Password Setup Input Before entering the settings interface, a password must be entered for verification to prevent unauthorized personnel from changing data and parameter settings.		N/A
18	Password Error If the password is entered incorrectly 3 times when accessing the settings interface, re-entry is required after 1 minute.		N/A

No.	Steps	Display Screen Indication	Sound
19	Settings Interface Configurable settings include: • Device name • Account and password settings • Preheating temperature settings • Ice crystal parameters settings • Protective temperature settings • Restore factory mode settings		N/A
20	Name and Account Modification • Device Type and MAC address cannot be modified. • To change the device name, administrator account, or operator account, enter the desired information in the respective fields under Settings. After completing all changes on the settings interface, click Confirm to save.		N/A
21	Temperature, Ice Crystal Parameters, and Password Modification Click the input fields for preheating temperature, ice crystal parameters, and password in the Settings menu to make changes. After completing all settings on the interface, click Confirm to save the changes.		N/A

PC Software Interface

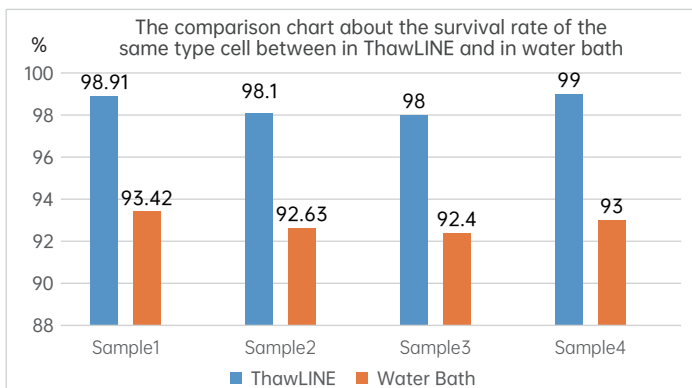
No.	Steps	Display Screen Indication	Sound
1	Check that the device is in good condition and connect the power supply.	N/A	N/A
2	Connect Device After starting the software, if the interface displays "Device Not Connected. Please Connect the Device!", please check the following: • Connect the cell thawing device to the computer via the data cable. • Once the system detects the connection, the prompt will disappear automatically.		N/A
3	Connect Port After a successful connection, select the corresponding port number and click Connect to enter the login interface.		N/A
4	Login to System Enter the password on the login interface and click the arrow on the right to log in. • If the password is incorrect, a login failure message will appear. • After successful login, you will enter the main interface.		N/A

No.	Steps	Display Screen Indication	Sound
5	Enter Main Interface The main interface shows the device status as "Device Connected", with four slots displayed below. • All slots currently show No Data, indicating that no experiment has been run or there is no historical data. • You can click any slot to view details.	 The screenshot shows the main interface with a green header. On the left, there's a 'Device Connected' status and a 'Test Record' table. In the center, there are four slots labeled 'Slot 1', 'Slot 2', 'Slot 3', and 'Slot 4', each with a 'No Data' indicator. On the right, there are two 'Temperature Curve' graphs showing data for Slot 1, Slot 2, Slot 3, and Slot 4.	N/A
6	No Records At this point, no experiment has started. Clicking Data Export will display No Records.	 The screenshot shows the main interface with a green header. A dialog box titled 'No Records!' is displayed in the center, with a 'OK' button. The background interface is the same as in step 5.	N/A
7	View Slot 1 Details After starting the experiment, click the first slot to enter the detailed information page, which includes the following modules: • Test Records: Displays all experiment records for this slot. • Device Information: Shows the module's sample batch number, device name, etc. • Experiment Data: Displays the data collected during the slot's operation. • Temperature Curve: Shows the temperature changes throughout the experiment.	 The screenshot shows the detailed information page for Slot 1. It includes a 'Test Record' table, a 'Device Information' section, an 'Experiment Data' section, and a 'Temperature Curve' graph. The 'Test Record' table shows a list of records with columns for 'Time', 'Temperature', and 'Humidity'. The 'Device Information' section shows details like 'Sample Batch Number', 'Device Name', 'Device ID', 'Device Type', 'Device Status', and 'Device Location'. The 'Experiment Data' section shows a list of data points with columns for 'Time', 'Temperature', and 'Humidity'. The 'Temperature Curve' graph shows the temperature changes over time.	N/A

No.	Steps	Display Screen Indication	Sound
8	Click the second slot; the content is the same as Slot 1, and other slots operate similarly. Click Change Password to open fields for New Password and Confirm Again, allowing you to modify the password.		N/A
9	Data Export After initiating the export, the system will display "Data Export in Progress". Do not close the program or disconnect the device during this process.		N/A
10	Export Complete After the export is finished, the system will display "Export Complete". The file will be saved to the default path.		N/A

Package Contents and Technical Specifications

List	Quantity	Notes
Cell resuscitator	1	Device
Power Cord	1	220V
Dedicated Data Cable	1	
SD Card	1	
Card Reader	1	
Dust Cover	1	
Temperature Probe	1	
Instrument manual	1	
Performance Parameters		
Thawing Time	2 to 8 minutes (varies depending on different sample containers)	
Rated Power	Maximum Power: 300W	
Input/Output Voltage (Power Adapter)	AC 100–240V / DC 12V, 50–60Hz	
Output Current	15A	
Power Plug	US,UK,EU	
Dimensions	L × W × H: 323mm × 288mm × 168 mm	
Certification	CE	
Resuscitation Method	Waterless Dry Heat Transfer	
Initial Temperature Protection	Yes, users can set it themselves.	
Temperature Calibration	Yes, with intelligent calibration for easy third-party measurement.	
One-Touch Sterilization	Yes, the instrument automatically sterilizes from the inside out.	
Ice Crystal Control	Customers can adjust the size of ice crystals after resuscitation according to their actual usage needs.	
Audit Trail	Supports connection to PC for exporting uneditable resuscitation temperature curve images.	
Permission Management	Level 2: Operator / Administrator	
Resuscitation Complete	With audible alerts, samples automatically eject.	
Display Screen	Touchscreen display allows real-time monitoring of the resuscitation chamber dynamics.	
Data Export	Data stored on SD card; export to PC via host computer connection.	



Safety Instructions

The automated cell thawing system is designed for use by well-trained and experienced laboratory personnel.

1. Please read the instructions before installing, using or cleaning the equipment;
2. To prevent risk of severe or fatal electric shock, use only the AC power supply provided with this package;
3. Do not place near or immerse in water or other liquids;
4. Unauthorized removal of the casing will void your valid warranty period. Do not remove the casing of the device unless authorized by Kemesser;
5. There is a jack on the back of the machine. Do not insert any plug into this port or attempt to connect the device to other devices (such as DC power supply, computer) unless authorized by Kemesser;
6. This device is designed for indoor benchtop laboratories. Make sure the device is safely placed on a level surface during use. It is not recommended to use it near strong vibrations such as centrifuges, shakers or vortex mixers;
7. Make sure the tubes are frozen in an upright and vertical position so that the frozen contents are at the bottom of the tube;
8. Make sure the cryogenic tubes have a clean appearance and that the caps are tightly sealed to prevent leaks;
9. Insert the cryogenic tube into the opening of the device and start the thawing process. Do not touch the tube with your hands or any other objects until the thawing process is completed;
10. The thawing time is generally 2-3 minutes. During use, laboratory personnel recommend staying close to the instrument so that it can be transferred immediately after thawing is completed;
11. Keep the device covered when not in use;
12. When the device is turned on, the heating element inside the vial port is heated. Avoid inserting body parts (such as fingers or any object other than the cryogenic vial) into the thawing port at any time.

Care and Cleaning

Do not attempt to clean the internal components. Do not insert any cleaning objects or solutions into the open thawing block on the top of the machine. Before or after each use, wipe the exterior with a soft, non-abrasive lab wipe moistened with 70% alcohol. Thawing condensation does not cause problems as it evaporates during the defrosting process or during standby.

Technical Service and Repair

Contact Kemesser technical support at 0571-87156759 or email andy@kemesser.com for help.

Certificate of Conformity:

Certificate of Conformity	
Name:	Automated cell thawing instrument
Model:	
Date:	
Test Result:	Qualified
Inspection date:	

Warranty Card:

Warranty:

1.Enjoy ONE year free warranty for the whole machine from the date of purchase;

2.The following situations are not covered by the warranty:

A: Failure and damage caused by not operating correctly according to the manual;

B: Failures and damages caused by self-disassembly and repair;

C: Failures and damages caused by human factors;

D: If the warranty card cannot be presented or the instrument does not have the official sales seal.

Warranty				
No.	Fault cause	Repair dare	Repair man	Remark
1				
2				
3				
4				
5				
6				
7				



KEMESSER
TECHNOLOGY

Address: Room 206, 14 Blocks, 168 Jianding Road, Shangcheng District,
Hangzhou City, Zhejiang Province, China.

Tel: 0571-87156759 87156259

Zip code: 310021

E-Mail: andy@kemesser.com rain@kemesser.com

Website: www.kemesser.com